



DORSAL PULVINAR CONTRIBUTIONS TO REACHES AND SACCADDES

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Summary

We examined how dorsal pulvinar population activity represents task variables during visually guided reach and saccade behavior. Neural activity was recorded while monkeys performed reach-only or saccade-only movements to peripheral targets, with task conditions varying by effector, reach hand, and target position. After preprocessing, population activity was analyzed using time-resolved decoding, temporal generalization matrix (TGM), and targeted dimensionality reduction (TDR). The results showed that target location was strongly represented after cue onset and remained decodable across the delay and movement periods, suggesting a stable spatial code. Timings of decodability from TGM were used to define TDR regressors. Reach-hand information was present even before cue onset, likely reflecting maintained hand posture at the central position, while effector-related information became strongest later in the trial in both the planning and movement period. TDR projections further showed that target position, hand context, and action-specific spatial variables structured dorsal pulvinar population activity along interpretable axes. Overall, the findings suggest that the dorsal pulvinar is not limited to visual or saccade-related processing, but participates more broadly in frontoparietal circuits linking spatial target information with effector and hand context during visually guided action.

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1 INTRODUCTION

During visually guided action, the brain must transform sensory information about a target, including its location in space, into a motor plan specifying the appropriate effector. This transformation is not simply a serial process from perception to movement, but can be understood as a competition between potential actions shaped by sensory evidence, behavioral goals, reward, and motivation (Cisek, 2006; Cisek & Kalaska, 2010; Leblois, Boraud, Meissner, Bergman, & Hansel, 2006; Pastor-Bernier & Cisek, 2011; Schultz, 2004). In reaching, the motor system must convert a spatial goal into a temporally structured pattern of arm and hand activity that brings the limb toward the target (Buneo, Jarvis, Batista, & Andersen, 2002; Cohen & Andersen, 2002; Georgopoulos, Kalaska, Caminiti, & Massey, 1982). Reaching behavior also depends on the spatial relationship between the acting hand and the target, and such relationships are reflected in cortical reach-related activity (Carey, Hargreaves, & Goodale, 1996; Chang, Dickinson, & Snyder, 2008; Fisk & Goodale, 1985; Heming, Cross, Takei, Cook, & Scott, 2019). Thus, visually guided action provides a useful framework for asking how neural populations jointly represent target location, effector choice, and movement context.

This introduction proceeds from general principles of neural representation to the specific circuit and method studied in this project. I first introduce mixed selectivity and population state-space approaches as a framework for understanding how neural activity can jointly represent multiple task variables. I then describe frontoparietal networks for eye-hand coordination and motivate a focus on the dorsal pulvinar, a thalamic structure connected with visual, parietal, and frontal cortical areas. Finally, I introduce targeted dimensionality reduction (TDR) as a method for testing whether target location, effector choice, and reach hand are represented along separable dimensions of pulvinar population activity.

1.1 *Mixed Selectivity*

Behavioral responses must be flexible across changing task demands, allowing animals to select appropriate actions while still generating movements rapidly in real time. A neural mechanism that can support this flexibility is mixed selectivity, in which individual neurons respond to combinations of multiple task variables rather than encoding one variable in isolation (Fusi, Miller, & Rigotti, 2016; Rigotti et al., 2013; Tye et al., 2024). For example, a neuron may not only be selective for target location, but may respond differently depending on the particular combination of target location, reach hand, effector, and task epoch.

Such selectivity can also be dynamic: the response properties of a neuron may change over time depending on the current task epoch and population state. Mixed selectivity has been described prominently in prefrontal, parietal, and motor-related areas, where multiple task variables can jointly influence neural activity and support flexible, context-dependent behavior (Diomedì, Vaccari, Filippini, Fattori, & Galletti, 2020; Mante, Sussillo, Shenoy, & Newsome, 2013; Rigotti et al., 2013). Although such responses can appear heterogeneous at the single-neuron level, they can form structured high-dimensional representations at the population level.

This high dimensionality gives mixed selectivity an important computational advantage. By representing different linear or nonlinear combinations of task variables, a

population can support a larger repertoire of input-output mappings and allow task-relevant variables to be separated by downstream readouts (Fusi et al., 2016). In prefrontal cortex, Rigotti et al. (2013) showed that multiple task variables could be decoded from population activity even when single-neuron selectivity was weak, and that error trials were associated with reduced representational dimensionality. Thus, the geometry of population activity can be directly related to behavioral performance.

Population activity can therefore be represented in a neural state space, where each axis corresponds to the activity of a single neuron and the temporal evolution of the population forms a trajectory through this space (Cunningham & Yu, 2014; Vyas, Golub, Sussillo, & Shenoy, 2020). Motor and premotor cortex provide useful examples of this framework: preparatory activity can place the population in an initial state appropriate for the upcoming movement, whereas movement-related activity evolves through dynamics associated with action execution (Ames, Ryu, & Shenoy, 2014; Churchland et al., 2012; Elsayed, Lara, Kaufman, Churchland, & Cunningham, 2016; Kaufman, Churchland, Ryu, & Shenoy, 2014). This framework makes it possible to ask not only whether neurons encode a task variable, but also whether variables such as target location, effector choice, reach hand, and task epoch are represented along separable dimensions of population activity.

1.2 Brain Networks for Eye-Hand Coordination

To guide action, the brain must identify behaviorally relevant targets and transform visual information into appropriate motor plans. This function has classically been associated with the dorsal visual stream, which projects from early visual cortex toward posterior parietal areas and supports the sensorimotor transformations required for visually guided behavior (Goodale & Milner, 1992; Milner, 2017). Within this pathway, posterior parietal cortex and frontal motor areas form frontoparietal networks for action planning (Andersen & Buneo, 2002; Battaglia-Mayer, Archambault, & Caminiti, 2006; Buneo & Andersen, 2006; Cui & Andersen, 2007; Wise, Boussaoud, Johnson, & Caminiti, 1997).

Posterior parietal cortex represents target location, gaze direction, spatial attention, and potential movement plans, whereas premotor and motor areas contribute to selecting and preparing appropriate actions (Andersen & Buneo, 2002; Buneo & Andersen, 2006; Churchland et al., 2012; Cisek, 2006; Paré & Wurtz, 2001; Snyder, Batista, & Andersen, 2000). These circuits are especially relevant for eye-hand coordination, because reaches are often planned and executed together with gaze shifts to the target. Parietal regions such as the parietal reach region (PRR) and lateral intraparietal area (LIP) have been implicated in linking visual target selection, saccade planning, and reach planning (Hwang, Hauschild, Wilke, & Andersen, 2014; Snyder et al., 2000). However, visually guided action is not controlled by cortical frontoparietal circuits alone, but emerges from distributed cortical-subcortical loops involving the basal ganglia, cerebellum, and thalamus (Hull, 2020; Klaus, Alves da Silva, & Costa, 2019; Popa, Streng, Hewitt, & Ebner, 2016). This distributed view motivates considering the pulvinar as a potential contributor to visually guided eye-hand coordination.

1.3 *Pulvinar Contributions to Action*

The pulvinar is the largest higher-order thalamic nucleus in the primate brain and has extensive reciprocal connections with visual, parietal, and frontal cortical areas (Arcaro, Pinsk, Chen, & Kastner, 2018; Arcaro, Pinsk, & Kastner, 2015; Grieve, Acuña, & Cudeiro, 2000; Shipp, 2003). Through these connections, it has been implicated in visual attention and in the selection of behaviorally relevant information (Cortes, Ladret, Abbas-Farishta, & Casanova, 2024; Shipp, 2003; Zhou, Schafer, & Desimone, 2016). The primate pulvinar is anatomically and functionally heterogeneous, consisting of multiple subdivisions with distinct connectivity patterns. In particular, the dorsal pulvinar is commonly considered to include the anterior pulvinar, medial pulvinar, and the dorsal part of the lateral pulvinar (Arcaro et al., 2015).

This organization is important for visually guided action. Ventral and inferior pulvinar regions are more strongly associated with occipital and temporal visual areas and are often discussed in relation to visual object processing and the ventral visual stream. By contrast, dorsal pulvinar regions show stronger coupling with posterior parietal and frontal cortex, placing them in a position to interact with dorsal-stream and frontoparietal circuits involved in spatial attention, target selection, and action planning (Arcaro et al., 2018; Cortes et al., 2024; Kaas & Lyon, 2007). Consistent with this view, lesion and inactivation studies suggest a causal role of the dorsal pulvinar in spatial attention and behavioral target selection, with disruption producing neglect-like impairments and deficits in selecting behaviorally relevant stimuli (Kaduk, Wilke, & Kagan, 2024; Wilke et al., 2018).

This raises the question of whether dorsal pulvinar population activity represents task variables relevant for visually guided action, such as target location, effector choice, and reach hand, and whether these representations change between movement preparation and execution. If dorsal pulvinar activity primarily reflects visual target processing, spatial variables should dominate its population activity. However, if the pulvinar participates more broadly in frontoparietal action networks, its population activity may also carry information about the upcoming effector, the hand used for reaching, or interactions between spatial and action-related variables. In this project, we therefore analyzed spiking activity recorded from monkeys performing an instructed reach/saccade task, in which trials differed in target location, effector choice, and, for reach trials, the hand used to perform the movement.

1.4 *Targeted Dimensionality Reduction*

To study task-related structure in pulvinar population activity, we used dimensionality reduction. Neural population activity is high-dimensional because each neuron contributes one dimension to the population state space. Dimensionality reduction methods identify lower-dimensional structure in these data and make it possible to visualize neural trajectories over time (Cunningham & Yu, 2014). Unsupervised methods such as principal component analysis (PCA) identify dimensions that explain the largest amount of overall variance, but these dimensions are not necessarily aligned with task variables of interest.

Targeted dimensionality reduction (TDR) provides a more direct way to study task-related representations. Instead of identifying dimensions only from the variance struc-

ture of the data, TDR defines axes based on experimentally specified variables. This approach has been used to study context-dependent computation in prefrontal cortex, where complex mixed single-neuron responses could be interpreted by projecting population activity onto task-relevant axes derived from regression coefficients (Mante et al., 2013). In this framework, each axis corresponds to the population activity pattern associated with a particular task variable, such as target location, effector choice, or reach hand.

In practice, we fit linear regression models relating each neuron’s firing rate to variables in the reach/saccade task. For a given task variable, the regression coefficients across neurons form a vector in population space, defining a TDR axis that captures the corresponding variable-related activity pattern. Because different task variables may produce correlated regression axes, the raw axes were orthogonalized before projection. We then projected condition-averaged population activity onto these axes. If a task variable is strongly represented in the population, conditions differing in that variable should separate along the corresponding axis.

TDR is related to other supervised or semi-supervised population methods, such as demixed principal component analysis (dPCA), which separates population activity into components associated with different task parameters (Kobak et al., 2016). However, TDR was better suited to the present project because it provides a small number of directly interpretable axes, with one axis corresponding to each task variable or interaction term.

The aim of this project was therefore to determine how dorsal pulvinar population activity represents visually guided actions. Specifically, we asked whether effector choice, reach hand, and target location are represented along separable dimensions of population activity, whether these representations change across task epochs, and whether combinations of task variables produce interaction-like structure in pulvinar state space. By projecting population activity onto TDR axes, we could examine not only whether these variables were represented, but also when each representation emerged during the trial.

2 METHODS

2.1 *Software and Code Availability*

All analyses were performed in Python 3 (Van Rossum & Drake, 2009). Data processing and statistical analyses were implemented using standard scientific Python packages, regression was performed with scikit-learn 1.9.0 (Pedregosa et al., 2011), and figures were generated with Matplotlib 3.10.3 (Hunter, 2007) and Plotly (Inc., 2015). Regression and . The analysis code is available at: https://github.com/dagdpz/dimred_neural_space.

2.2 *Experimental Task and Recordings*

Neural activity was recorded from the dorsal pulvinar of two monkeys, F and L, while they performed a visually guided reach/saccade task. Trials required either a saccade, an arm movement, or a combined eye-saccade-reach movement toward one of six peripheral target locations. The task varied three main behavioral factors: instructed effector, reach hand, and target position.

Each trial began with central eye fixation and placement of one finger on the central position of a touchscreen. Maintaining both eye fixation and hand position was required for the trial to proceed. A visual cue then instructed the movement type. After a fixed delay of 1.3 s from cue onset, a go signal indicated that the monkey could execute the instructed movement. During the delay period, the monkey maintained central fixation and hand position. On reach trials, the monkey reached to the target using the hand that had been held at the central position. On saccade trials, the monkey made an eye movement to the target while maintaining the required hand position. Task conditions were interleaved across trials, so the upcoming effector and target position could not be predicted before cue presentation. For the analyses reported here, only reach-only and saccade-only trials were included. Combined reach-saccade trials were excluded from the main analysis.

2.3 *Preprocessing*

The two monkeys were analyzed separately. The data were stored in MATLAB files containing unit-level spike times and trial-level behavioral metadata, including task condition labels and event timings. Only accepted trials were included, where acceptance was based on signal-to-noise criteria during spike sorting. In addition, only successful trials of the relevant task type were retained.

Behavioral labels were recoded relative to the recorded pulvinar hemisphere. Reach hand was expressed as ipsilateral or contralateral to the recording side. Target position was also expressed in hemisphere-relative coordinates: for right-hemisphere recordings, the horizontal target coordinate was sign-flipped so that contralateral target positions were always positive and ipsilateral target positions were always negative. This ensured that spatial variables had the same interpretation across hemispheres and monkeys.

Spike times were converted into spike-density functions (SDFs). For each unit, spikes were binned at 1 ms resolution and smoothed with a Gaussian kernel to produce a continuous firing-rate estimate in Hz. Units with very low activity were excluded by

requiring a mean raw firing rate of at least 1 Hz. Firing-rate traces were then square-root transformed before analysis. This transformation is commonly used for Poisson-like spike-count data because it reduces the dependence of variance on the mean and makes responses more compatible with linear-Gaussian analyses (Anscombe, 1948; Lee, Port, Kruse, & Georgopoulos, 1998).

For the main cue-aligned analysis, neural activity was extracted from -0.5 to 2.0 s relative to cue onset. Cue onset, go signal, movement onset, and movement end were extracted from the behavioral state sequence for each trial. Event times other than cue onset were expressed relative to cue onset, so that cue onset occurred at time zero for all trials. A unit was included only if it had at least 5 trials in each condition.

2.4 *Time-Resolved and Temporal Generalization Decoding*

Time-resolved decoding was used to test whether task variables could be read out from population activity at different times during the trial. This analysis complemented TDR: whereas TDR defines interpretable task-related axes in neural state space, decoding tests whether population activity contains information sufficient to distinguish task conditions.

Decoding was performed using the same cue-aligned neural responses used for TDR. At each time point, pseudo-population activity vectors were constructed by combining activity across recorded units. Each pseudo-trial contained one activity value per unit, and the class label corresponded to the task variable being decoded. Separate binary decoders were trained for instructed effector, target space, and reach hand.

Real trials were split into training and test sets separately for each unit and class using five-fold cross-validation. Pseudo-populations were then generated independently from the training and test pools, ensuring that the same real trial could not contribute to both training and test data. For each class, 200 pseudo-trials were sampled per fold. At each time bin, a logistic regression classifier was trained to predict the task label from the population activity vector. Decoding accuracy was averaged across folds, and the standard error of the mean was computed across folds.

Temporal generalization decoding was used to examine whether neural codes were stable across time (King & Dehaene, 2014). A classifier was trained at one time point and tested at all other time points. Repeating this procedure for all training times produced a temporal generalization matrix, in which rows corresponded to training time and columns to testing time. Above-chance accuracy away from the diagonal indicates that a code learned at one time point generalized to other parts of the trial. Broad square-shaped regions of above-chance accuracy were interpreted as evidence for relatively stable population codes.

2.5 *Regressor Definition for TDR*

Task variables were converted into numerical regressors for targeted dimensionality reduction. The aim was to define interpretable axes in population activity space corresponding to condition-independent task epochs, spatial target position, instructed effector, reach hand, and their interactions.

First, four condition-independent regressors were defined to capture activity changes common across task conditions. These regressors were binary time-dependent vectors

with the same length as the neural response vector. The cue regressor, *cueCI*, was active from 50 to 200 ms after cue onset. The planning regressor, *planCI*, was active from 200 ms after cue onset until the go signal. The go regressor, *goCI*, was active from 50 to 200 ms after the go signal. The movement regressor, *movCI*, was active from movement onset for 300 ms. These regressors captured task-epoch-related activity that was not specific to effector, hand, or target condition.

Reach hand was encoded relative to the recorded hemisphere, with contralateral reach hand coded as +1 and ipsilateral reach hand coded as -1. Target position was represented using continuous two-dimensional target coordinates. The horizontal coordinate was used as *space_x*, and the vertical coordinate was used as *space_y*. To isolate target-related activity around cue presentation, these spatial regressors were multiplied by the cue-period mask, producing *space_x_cue* and *space_y_cue*.

Effector-related regressors were defined separately for saccade and reach trials. Saccade trials were coded with a saccade regressor, whereas reach trials were separated into *ipsi_hand* and *contra_hand* regressors. To capture action-specific spatial tuning, these action regressors were multiplied by the continuous target-position regressors. This produced action-by-space regressors such as *saccade_space_x*, *saccade_space_y*, *ipsi_hand_space_x*, and *contra_hand_space_y*.

Action-by-space regressors were further separated into preparatory and movement-related components. During the delay period, they were multiplied by the planning mask, producing regressors with the suffix *plan*, such as *saccade_space_x_plan*. During the movement period, the same action-by-space terms were multiplied by the movement mask, producing regressors with the suffix *mov*. This allowed separate population axes to be estimated for preparatory and movement-related spatial representations.

2.6 Targeted Dimensionality Reduction

For each unit, a separate linear regression model was fit between the square-root-transformed firing-rate response and the full set of task regressors. The response vector consisted of the unit's activity across all trials and time points in the analysis window. The fitted regression coefficients quantified how strongly each unit was modulated by each task variable. For each regressor, coefficients were then stacked across units to form a population vector in neural state space.

The raw regression vectors were orthogonalized using Löwdin symmetric orthogonalization (Löwdin, 1950). This produced an orthonormal set of TDR axes while preserving the original regression vectors as closely as possible. Condition-averaged population trajectories were then projected onto these axes to visualize how neural activity evolved over time with respect to cue-related, hand-related, spatial, preparatory, and movement-related task variables. Separation between condition-averaged trajectories along a given axis was interpreted as evidence that the corresponding task variable was represented in the dorsal pulvinar population.

As control, we repeated the TDR analysis using a train/test split. Trials were split separately within each unit and condition, so that each unit contributed trials from every condition to both the training and test sets. 70% of trials were assigned to the training set and the remaining 30% were held out for testing. TDR axes were defined only from the training trials. Condition-averaged trajectories were then computed from the held-out test trials and projected onto the train-defined axes. This procedure tested whether the

task-related axes generalized to independent trials, rather than merely describing the same data used to estimate the regression coefficients. To assess the stability of the held-out projections, the train/test split was repeated across random partitions, and projection time courses were summarized as the mean $\pm$ SEM across splits.

3 RESULTS

3.1 Data Description

After preprocessing and filtering, the dataset contained a total of 276 recorded units across the two monkeys. Monkey F contributed 201 units and 38,845 unit-trial observations. On average, each unit from Monkey F contributed 193.26 trials (SD = 52.17). Monkey L contributed 75 units and 15,726 unit-trial observations, with an average of 209.68 trials per unit (SD = 35.43). Because Monkey F provided the larger dataset, all subsequent analyses and plots shown in this section were performed on Monkey F.

Before TDR, spike-density functions were transformed using a square-root transform. Before transformation, firing rates were concentrated near low values with a long right tail extending to high firing rates. After the square-root transform, the distribution became more compressed and less dominated by extreme high-rate values, while preserving the non-negative structure of the data (Figure 1). This transformed activity was then used as the input for subsequent decoding and TDR analyses.

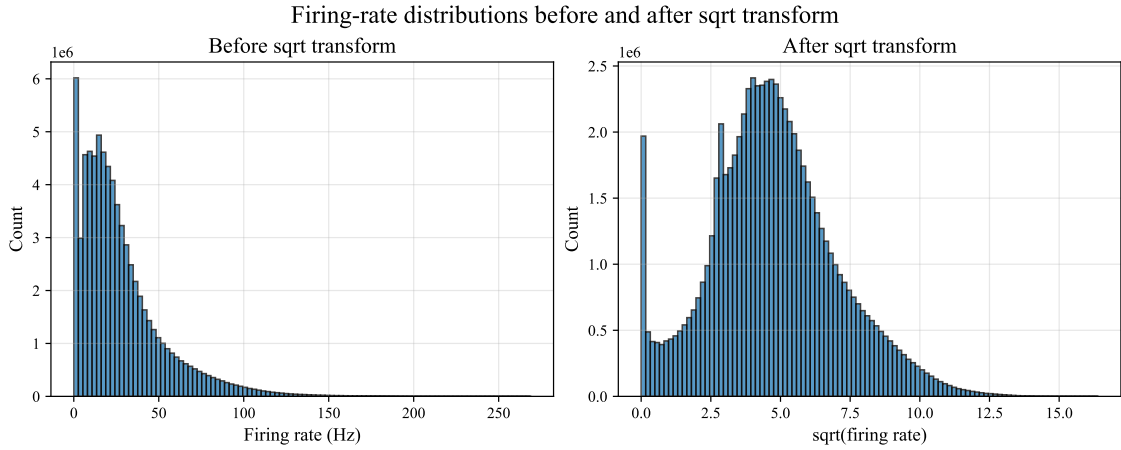


Figure 1: Firing rate distribution before and after square root transformation.

3.2 Temporal Generalization Matrices

Temporal generalization decoding revealed reliable information about target hemifield in dorsal pulvinar population activity (Figure 2). Decoding accuracy was close to chance before cue onset, as expected, because only central fixation was present and the upcoming target location was not yet specified. After cue onset, decoding accuracy increased rapidly, indicating that spatial information became available shortly after target presentation.

The TGM revealed four broad temporal regimes. First, pre-cue activity contained little decodable spatial information. Second, immediately after cue onset, spatial decoding increased sharply, but this early cue-related activity generalized only weakly to later test times, suggesting a transient cue-evoked spatial response. Third, during the subsequent delay/planning period, decoding generalized across a broader range of train and test times, indicating a more stable representation of target hemifield. This generalization extended beyond the GO signal, suggesting that spatial information persisted into later

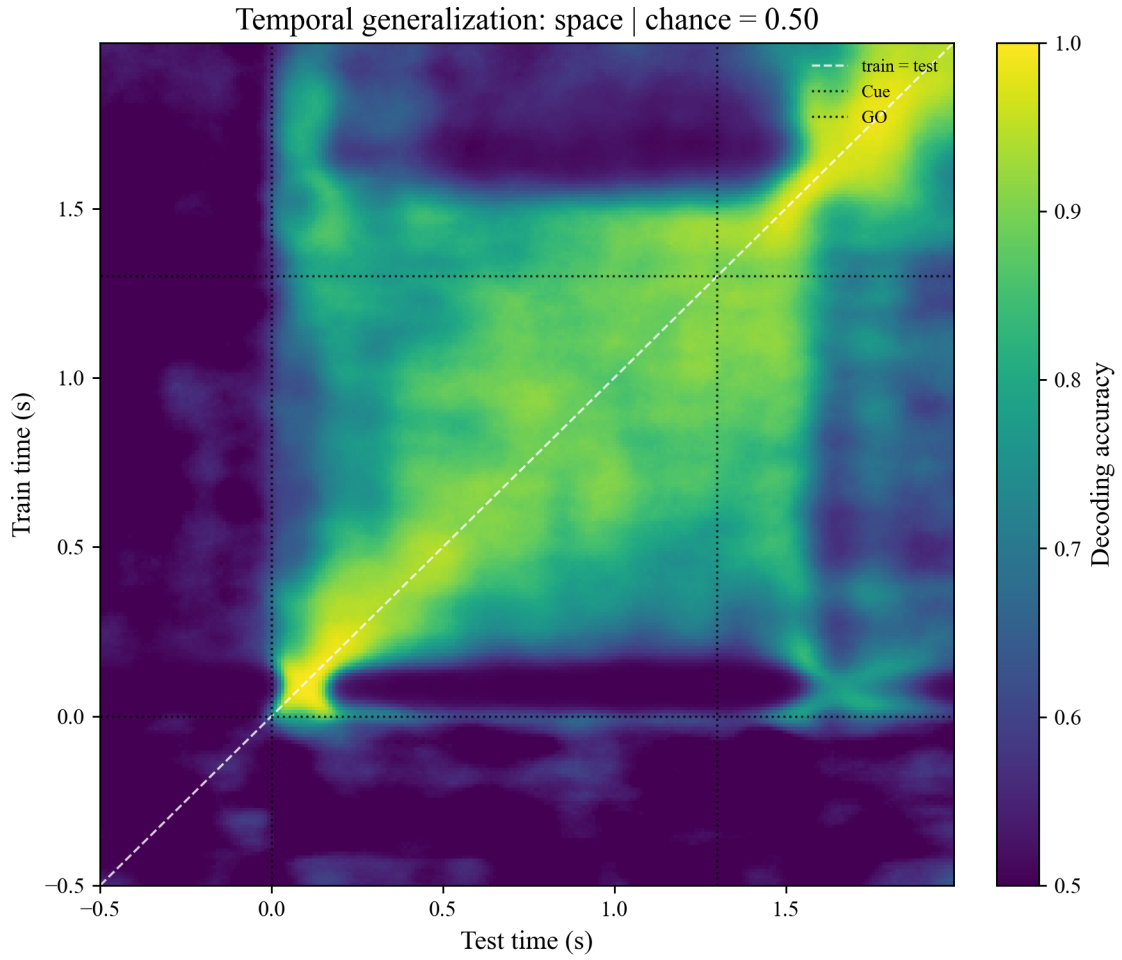


Figure 2: Temporal generalization matrix for horizontal target position decoding.

task phases. Fourth, around the movement period, spatial decoding again showed strong temporal generalization, consistent with a movement-related spatial population state.

These temporal regimes were used to guide the epoch definitions for the TDR analysis. In particular, the early cue period was used to define cue-related spatial axes, the delay period was used to define planning-related axes, and the late post-GO/movement period was used to define movement-related axes. Thus, the TDR windows were not chosen arbitrarily, but were motivated by the temporal structure observed in the decoding generalization analysis.

3.3 Cue-period representation of target position

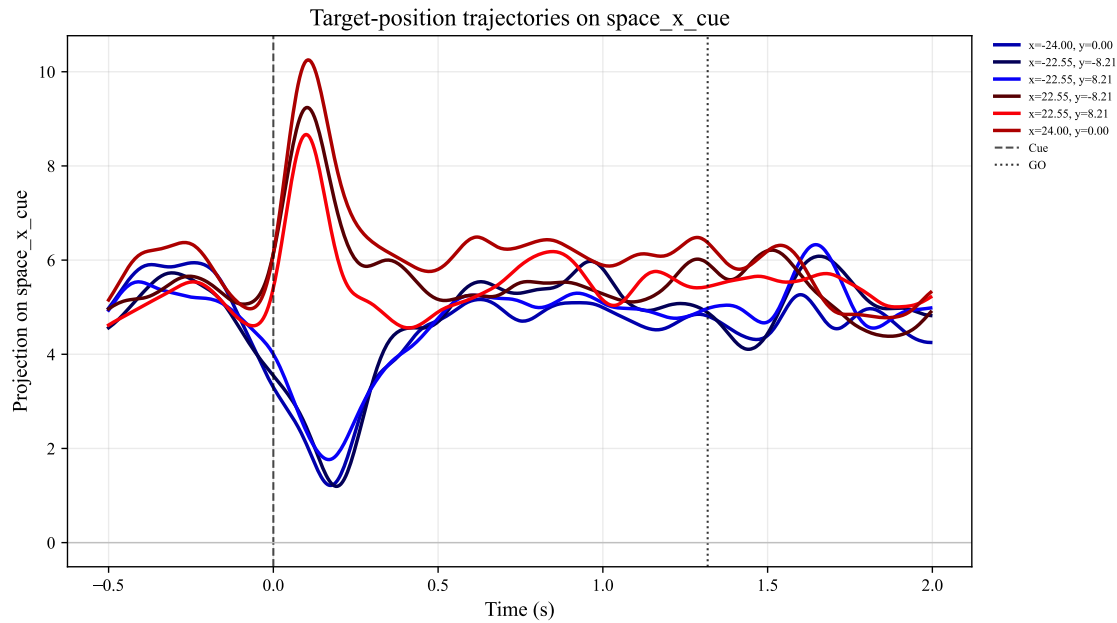


Figure 3: Cue-period representation of horizontal position.

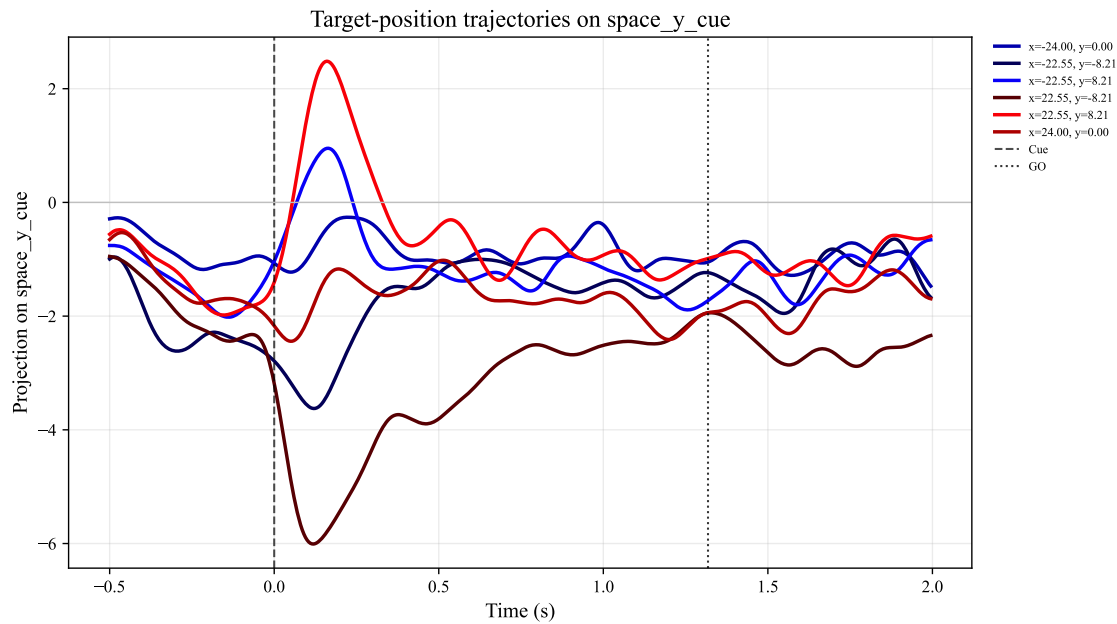


Figure 4: Cue-period representation of vertical position.

To examine whether the pulvinar population activity contained information about the instructed target location itself during the cue period, we projected condition-averaged neural trajectories for the 6 locations of the target onto two TDR axes: *space_x_cue* and *space_y_cue* (Figure 3 and Figure 4). These plots suggest that dorsal pulvinar population activity contains a cue-locked representation of target position in both directions. The y-position plots also show a clear continuous gradation as it includes the zero point. The y-position plot shows that the activity is only separable for targets that are contralateral

in horizontal position. The cue-period spatial axes showed strongest structure around cue onset, whereas later activity appeared to move into other task-related dimensions.

3.4 Hand representation

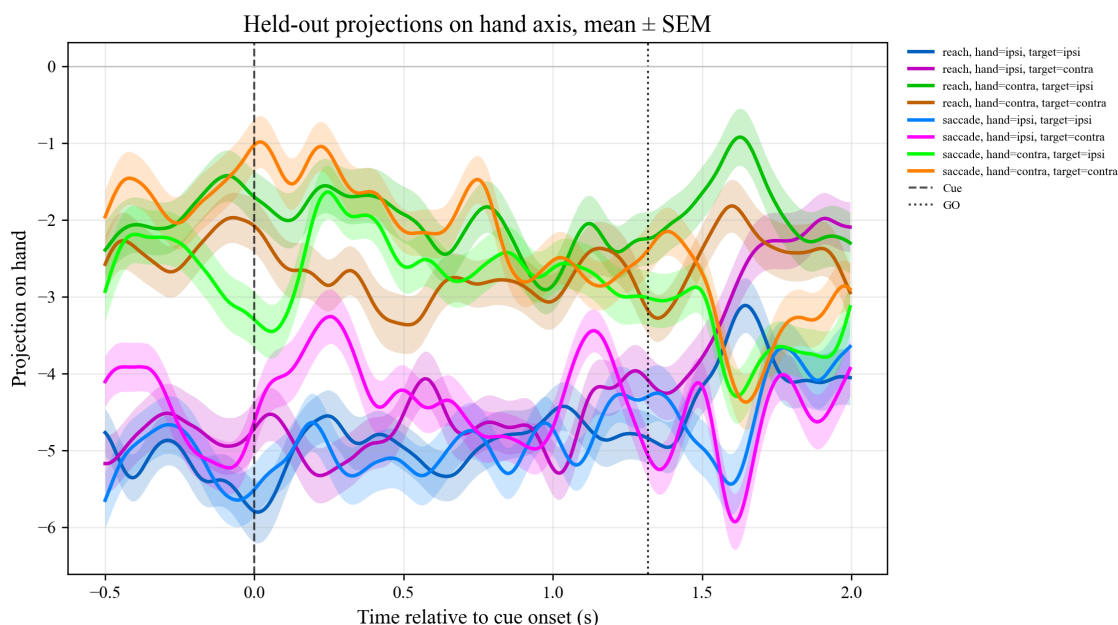


Figure 5: Hand representation.

Projection onto the train-defined hand axis revealed a clear separation between ipsilateral- and contralateral-hand conditions in held-out test trials. This separation was present not only during the delay and movement periods, but also during the pre-cue baseline period. Shaded regions indicate the SEM across repeated random train/test splits, and the same convention is used for all subsequent held-out projection plots.

Importantly, the separation was also visible in saccade trials, even though no reach movement was instructed on those trials. This suggests that the hand axis may not exclusively reflect reach planning, but may instead capture the maintained hand context or posture required at the beginning of each trial. Because the monkey had to keep one hand at the central position before cue onset, hand-related information was already available before the instructed movement was specified. Later modulation of the trajectories around the go cue and movement period may reflect a transition from maintained hand posture to movement execution or more general action-related processing.

3.5 Saccade representation during Planning and Movement

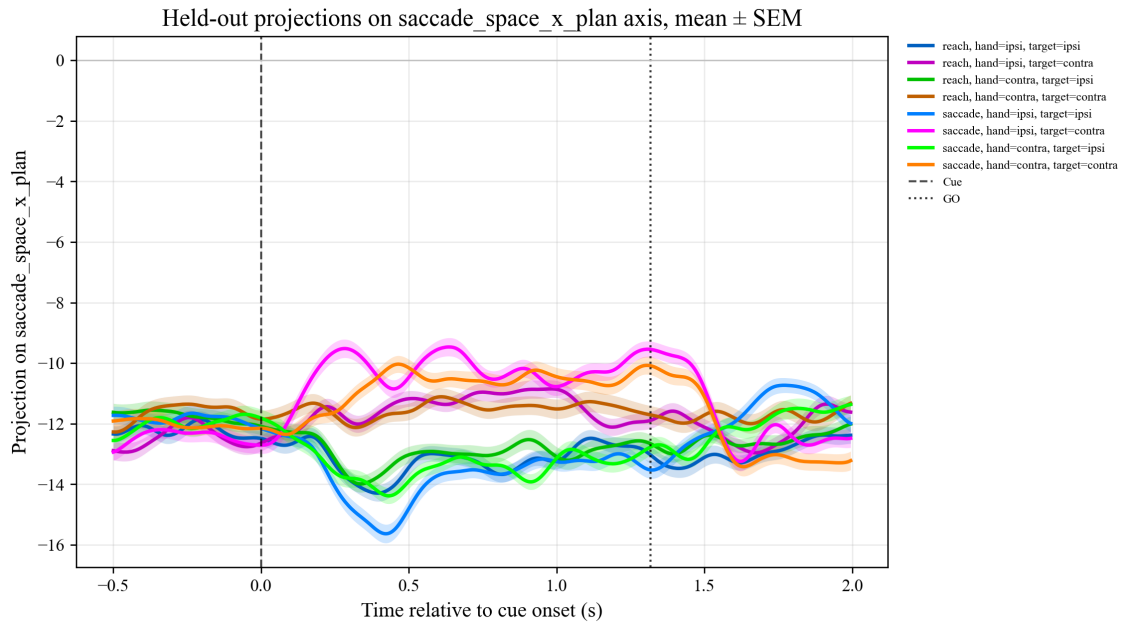


Figure 6: Saccade horizontal representation during planning.

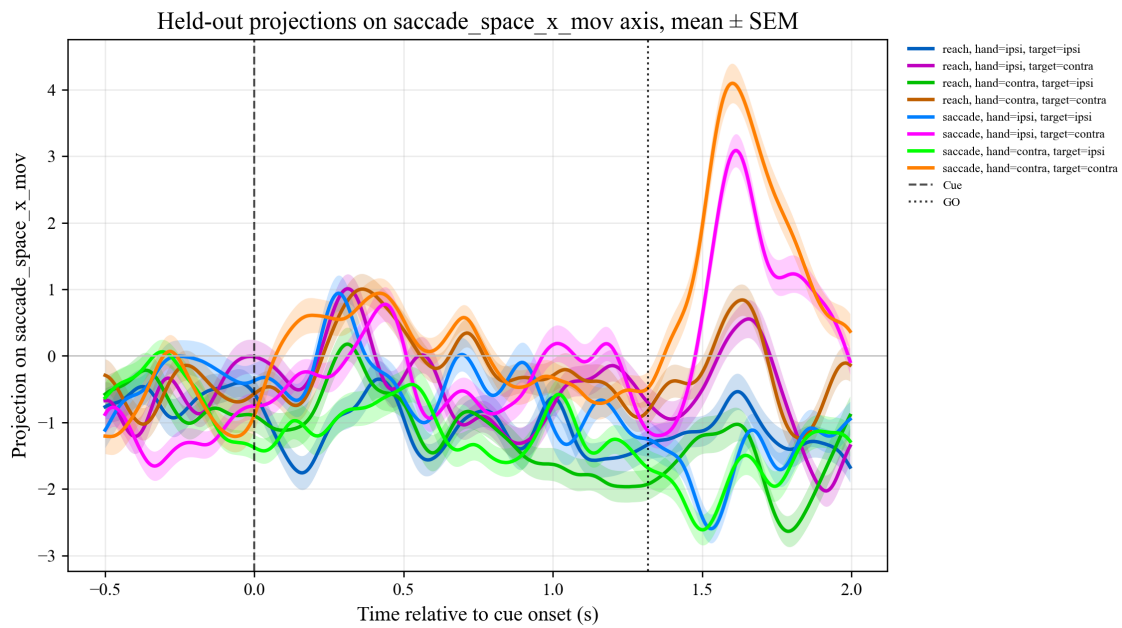


Figure 7: Saccade horizontal representation during movement.

We next examined whether horizontal spatial information was expressed differently during saccade planning and movement epochs (Figure 6 and 7). Condition-averaged trajectories were projected onto two axes defined by the interaction between saccade trials and horizontal target position: one defined during the planning period (saccade_space_x_plan) and one defined during the movement period (saccade_space_x_mov). Projection onto the planning-period axis revealed structured separation between ipsi-lateral and contralateral target conditions, indicating that dorsal pulvinar population

activity contained horizontal target-related structure along a saccade-defined planning dimension.

Projection onto the movement-period axis also showed target-related structure, but the pattern differed from the planning-period projection. Thus, horizontal target information was expressed along both saccade-related axes, but the corresponding population dimensions were not identical. This suggests that target-location information in dorsal pulvinar may be embedded in different action-context dimensions across the task. Because these axes were defined using epoch-specific regressors, these plots should be interpreted as descriptive visualizations of task-related population structure rather than as independent evidence for the precise timing of the spatial signal.

3.6 *Hand representation during Planning and Movement*

We next examined whether horizontal target information was expressed along hand-specific spatial axes (Figure 8 to 11). Condition-averaged trajectories were projected onto axes defined by the interaction between reach hand and horizontal target position, separately for ipsilateral- and contralateral-hands. Projections in Figure 8 and 10 showed structured separation between target hemifields during the planning period which even extended into the movement after the GO signal. This is similar to what was seen in saccades.

Importantly, this structure was not restricted to reach trials. Saccade conditions also separated along these hand-defined axes, suggesting that the axes are not purely hand-specific. Instead, they may capture a spatial target representation that is partly shared across effectors, or reflect the common visual target structure present across both reach and saccade trials. Therefore, these projections should be interpreted as evidence for structured spatial modulation along hand-defined axes, rather than definitive evidence for a hand-exclusive spatial code.

Together, these results suggest that horizontal target position is represented in dorsal pulvinar population activity in a way that can be expressed within multiple action-context dimensions.

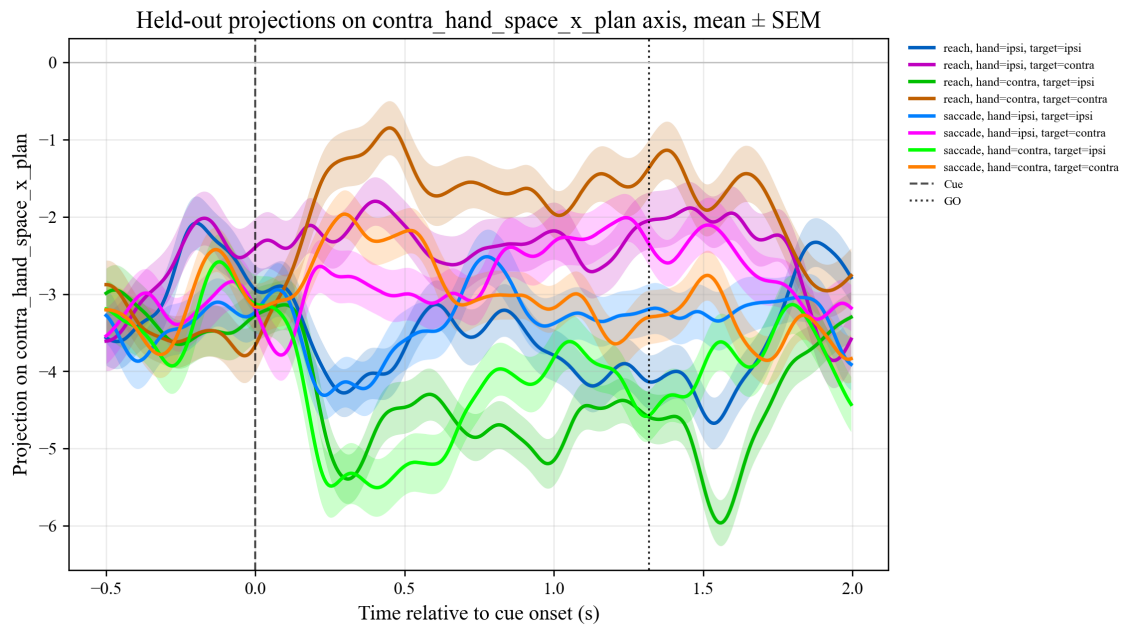


Figure 8: Contralateral hand planning representation

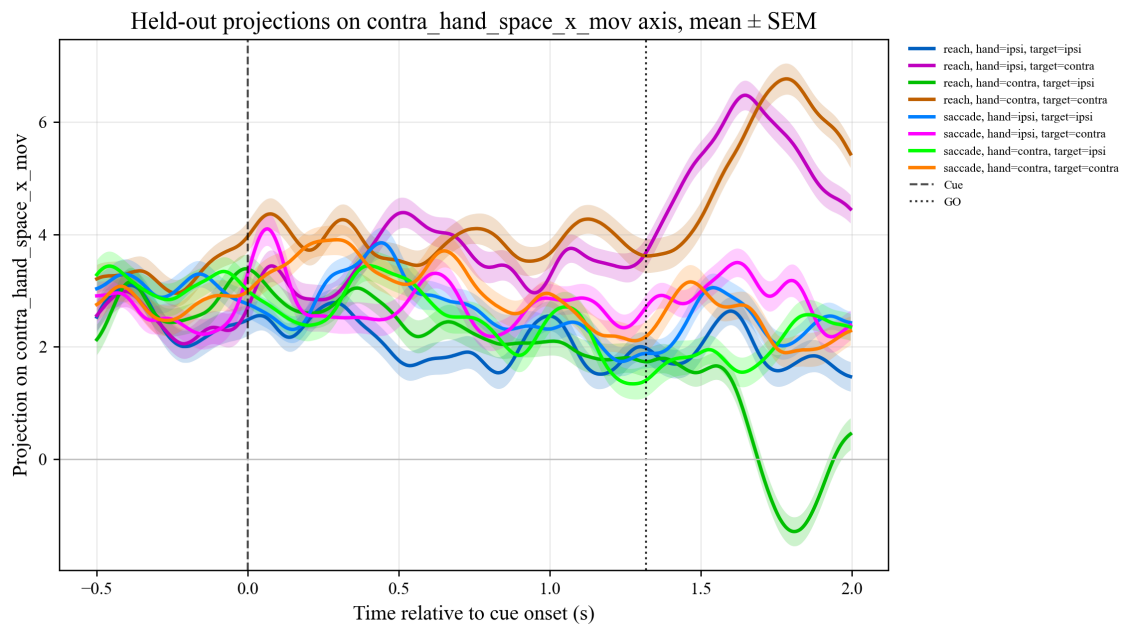


Figure 9: Contralateral hand movement representation

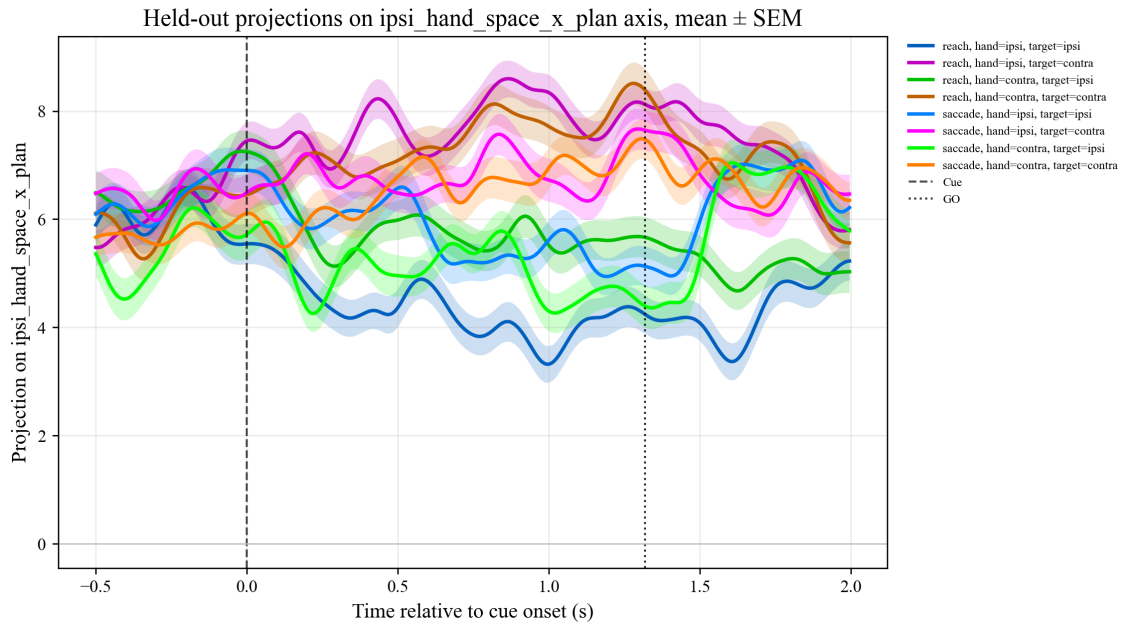


Figure 10: Ipsilateral hand planning representation

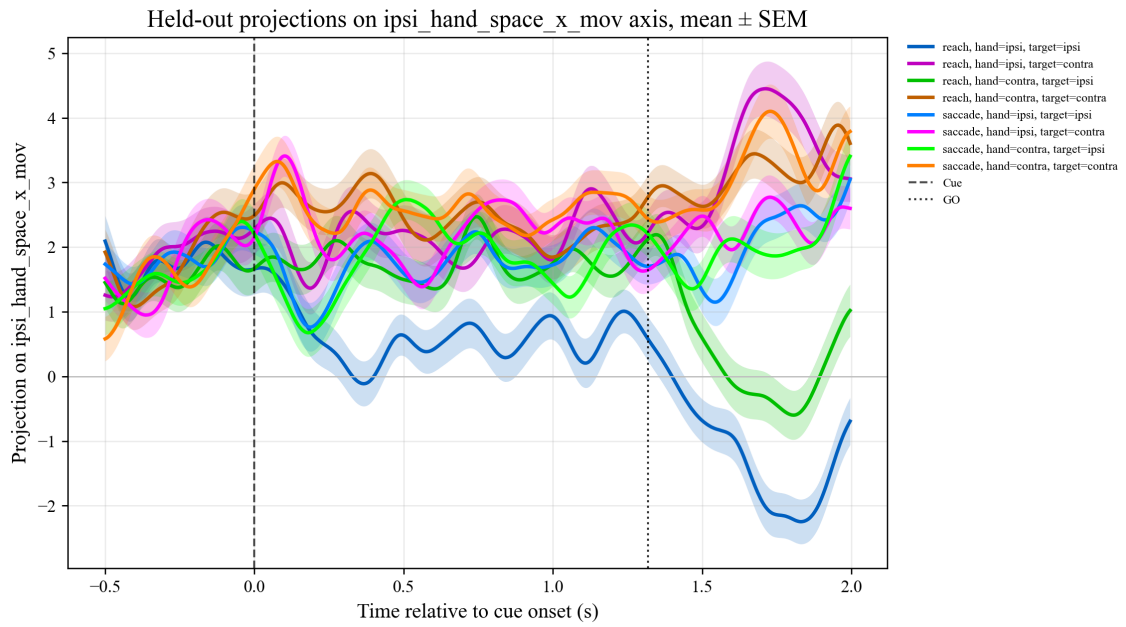


Figure 11: Ipsilateral hand movement representation

4 DISCUSSION

The aim of this project was to examine how dorsal pulvinar population activity represents task variables during visually guided reach and saccade planning. Using time-resolved decoding, temporal generalization, and targeted dimensionality reduction, we found that dorsal pulvinar activity contained reliable information about target location, effector context, and reach hand. Horizontal target position was the most robustly represented variable: target hemifield could be decoded shortly after cue onset, and cue-defined spatial axes separated ipsilateral and contralateral targets. This is consistent with previous work linking the pulvinar to spatial attention and target selection (Cortes et al., 2024; Shipp, 2003; Wilke et al., 2018; Zhou et al., 2016).

Importantly, the results suggest that dorsal pulvinar activity is not restricted to saccade-related spatial processing. Although the pulvinar is often discussed in relation to visual attention, spatial selection, and eye movements, the present analyses also revealed structure related to reach hand and hand-dependent target position. Projection onto the hand axis separated ipsilateral- and contralateral-hand conditions, and hand-defined spatial axes showed systematic target-related structure. This suggests that dorsal pulvinar activity contains information relevant for visually guided reaching as well as saccades.

These findings support a broader view of the dorsal pulvinar as part of an eye–hand coordination network. In the present task, the same visual target could guide a saccade, a reach with either hand, or potentially a coordinated eye–hand action. Dorsal pulvinar activity may therefore help maintain and route spatial information in a format that can be flexibly linked to different effectors. Rather than representing target location only in a saccade-centered format, pulvinar population activity appears to combine target location with action and hand context. This is consistent with the anatomical position of dorsal pulvinar within frontoparietal networks involved in spatial attention, target selection, and visually guided reaching (Andersen & Buneo, 2002; Arcaro et al., 2018; Buneo & Andersen, 2006; Cortes et al., 2024; Snyder et al., 2000).

The TDR results should nevertheless be interpreted cautiously. Because the axes were defined using task regressors and epoch-specific masks, the projection plots are best understood as descriptive visualizations of task-related population structure. They show how condition-averaged activity is organized relative to predefined task axes, but do not by themselves prove that these axes are used by downstream circuits. As an initial control, we repeated the TDR analysis after shuffling condition labels. These shuffled controls produced random-looking trajectories without consistent condition separation, suggesting that the structure observed in the real projections was not simply a consequence of the projection method itself.

Further statistical controls will be important to quantify these effects more rigorously. Future analyses should compare the real condition separations against a null distribution obtained from repeated label shuffling or permutation tests. Separation between conditions could be quantified as the distance between projected trajectories along each TDR axis, averaged within predefined task epochs. This would allow direct statistical testing of whether target, hand, or effector separations exceed chance. Cross-validated TDR would provide an additional control, by fitting axes on one subset of trials and testing whether the same axes separate held-out trials. Together, these analyses would help distinguish reliable task-related population structure from separation that could arise from overfitting or circular axis definition.

A further limitation is that the present analysis focused only on dorsal pulvinar. Because dorsal pulvinar is embedded in a larger frontoparietal network, it remains unclear whether the observed task-related dimensions originate within pulvinar, are inherited from cortical inputs, or reflect recurrent thalamo-cortical interactions. Future studies could address this by recording simultaneously from dorsal pulvinar and connected cortical areas. Communication-subspace analyses could then test whether the spatial and hand-related dimensions identified here are transmitted to parietal or frontal cortex, or whether they remain local to pulvinar population activity (Semedo, Zandvakili, Machens, Yu, & Kohn, 2019).

Causal experiments will also be important. The present results show that task variables can be decoded from dorsal pulvinar activity, but decoding does not establish that this activity is necessary for behavior. Reversible inactivation or stimulation of dorsal pulvinar during specific task epochs could test whether pulvinar activity is required for target selection, effector choice, hand-specific reach planning, or eye–hand coordination. Such experiments would extend previous lesion and inactivation work linking pulvinar to spatial attention and target selection (Kaduk et al., 2024; Wilke et al., 2018).

In summary, dorsal pulvinar population activity contained structured representations of target location and action context during visually guided behavior. While horizontal target position was most strongly represented, the population structure was not limited to saccade-related dimensions. Hand-related and hand-by-space axes also showed systematic condition structure, suggesting that dorsal pulvinar contributes to visually guided reaching as well as saccades. Shuffled-label controls supported this interpretation by disrupting the structured trajectory separation observed in the real data. Future statistical analyses using permutation testing, cross-validated TDR, and trajectory-distance measures will be important for quantifying the reliability of these effects. Overall, these findings support the view that the dorsal pulvinar participates in distributed frontoparietal circuits where spatial information is flexibly linked to different effectors and body contexts.

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